Beth Pearson

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SUMMARY

I build evaluation methods and interpretability tools tailored to the subtle failure modes of Vision-Language Models, specifically focusing on compositional generalisation, negation, and spatial reasoning. I also have previous experience building products for practical applications, having worked in production web-app development and building proof-of-concept LLM tools for applications in radiology. I'm looking to apply my background in machine learning and software development to solve practical, real-world problems.
Languages: Python (Primary), Bash, PHP, MySQL, MATLAB
Technologies: PyTorch, scikit-learn, HuggingFace
Tools & Infrastructure: Git, Slurm / GPU Clusters, WandB, Linux

PROFESSIONAL EXPERIENCE

AI Research Intern Sep 2024 – Nov 2024
Rosenfield Health Tech Ltd, Cardiff, United Kingdom

- Developed an NLP system in **Python** using clinical Named Entity Recognition and local LLMs to compare preliminary and final radiology reports, identifying clinically meaningful differences to support radiologist training and feedback
- Evaluated **Llama and Mistral** models for radiology report comparison, comparing their performance against radiologist-provided ground truth to identify the most effective approach.
- Worked cross-disciplinarily with **radiologists and researchers** to develop a proof-of-concept MVP, translating clinical requirements into a practical system that met the needs of its intended users while meeting the required privacy constraints

Software Developer Jun 2020 – Jul 2022
Wilxite, Stroud, United Kingdom

- Maintained and extended a production **PHP/JavaScript invoicing and client-management system** for a flood restoration company, developing new website features and automating previously manual spreadsheet-based workflows
- Developed and maintained **MySQL databases**, including building database structures from test data for a new client website developed from scratch
- Worked directly with clients to understand their business processes and requirements, translating these into practical software features and user-facing functionality
- Contributed to **system administration and infrastructure maintenance**, including Linux system updates, server backups and recovery, and managing server configuration and information using **Puppet**

Software Engineering Intern Jun 2019 – Sep 2019
Oracle, Bristol, United Kingdom

- Built an internal web app (Java, Vue.js) and added Prometheus/Grafana monitoring for Oracle Cloud Functions; deployed with Docker

ACADEMIA

Interactive Artificial Intelligence, PhD Sep 2022 - Nov 2026
Department of Computer Science, University of Bristol
Thesis Title: Understanding Compositional Reasoning in Vision-Language Models

- Built workflows using PyTorch and HuggingFace to analyse Vision-Language Models (VLMs) such as PaliGemma, LLaVA, and InternVL, identifying specific failure modes in spatial reasoning, attribute binding, and negation
- Designed novel evaluation benchmarks to systematically stress-test VLM reasoning capabilities beyond standard leaderboards
- Managed distributed training and evaluation jobs across HPC / multi-GPU clusters

Engineering Mathematics, MEng (First Class Honours) Sep 2016 - Jun 2020
Department of Engineering Mathematics, University of Bristol

- **Final Year Project – Activity Recognition from LiDAR Sensing in Smart Homes:** Built ML models to classify human activities from raw point cloud data. Performed unsupervised segmentation on LiDAR sensor data, comparing the performance of different neural networks on correctly identifying human activities such as walking and sitting

SELECTED PUBLICATIONS

[1] Beth Pearson, Michael Wray, and Martha Lewis. 2026. Probing Negation Understanding in Vision-Language Models With Object Attention. In *Findings of the Association for Computational Linguistics: EMNLP 2026*,, Budapest, Hungary. Association for Computational Linguistics.

[2] Beth Pearson, Bilal Boulbarss, Michael Wray, and Martha Lewis. 2025. Evaluating Compositional Generalisation in VLMs and Diffusion Models. In *Proceedings of the 14th Joint Conference on Lexical and Computational Semantics (*SEM 2025)*, Suzhou, China. Association for Computational Linguistics.

[3] Pearson, B., Adnan, A., Abdallah, Z.S. (2026). Semantic Similarity in Radiology Reports via LLMs and NER. Artificial Intelligence for Biomedical Data. AIBIO 2025. Communications in Computer and Information Science. https://doi.org/10.1007/978-3-032-17216-7_10

AWARDS, TEACHING AND SERVICE

- **Awards & Talks**: Best Poster Award (UK AI 2025); oral presentation, ECAI 2025; invited talk, UCL (2025).
- **Teaching Assistant:** Supported undergraduate Computer Science and Mathematics modules, including supervising student projects for Intro to AI (2024).
- **Peer Reviewer:** Reviewed conference and journal papers for ACL Rolling Review and IET.

ADDITIONAL INFORMATION

- **Driving Licence:** Full UK Driving Licence
- **Interests:** Running, Rock Climbing, Playing Keyboard, Live Music